An Adaptive and Explainable ML-Based Intrusion Detection System for Evolving Cyber Threats

A Capstone Project Abstract

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ABSTRACT

Conventional Machine Learning-based Intrusion Detection Systems (IDS) often rely on static models, making them less effective against evolving and previously unseen cyber threats. This project proposes an adaptive IDS that integrates Cyber Threat Intelligence (CTI) with hybrid machine learning and online learning. The system uses XGBoost as the supervised classifier and Isolation Forest for anomaly detection, enabling identification of known malicious traffic and uncertain network behavior. When anomalous or low-confidence traffic is detected, relevant Indicators of Compromise (IoCs) are extracted and submitted to CTI sources for additional threat information. The CTI Transfer Model processes the obtained intelligence and correlates it with network sightings to generate validated training data. Concept drift detection identifies changes in network traffic patterns and triggers model updates when required. The updated data is incrementally incorporated into the IDS, enabling continuous adaptation to emerging attacks. SHAP-based Explainable AI identifies important features influencing predictions. The framework aims to improve detection accuracy, adaptability, and interpretability while reducing the limitations of static IDS models.

Keywords: Intrusion Detection System (IDS), Machine Learning, Cyber Threat Intelligence (CTI), XGBoost, Isolation Forest, Online Learning, Concept Drift Detection, Explainable AI, SHAP, Indicators of Compromise (IoCs).

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Y. -D. Lin, Y. -H. Lu, R. -H. Hwang, Y. -C. Lai, D. Sudyana and W. -B. Lee, "Evolving ML-Based Intrusion Detection: Cyber Threat Intelligence for Dynamic Model Updates," IEEE Transactions on Machine Learning in Communications and Networking, vol. 3, pp. 605-622, 2025, doi: 10.1109/TMLCN.2025.3564587.

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